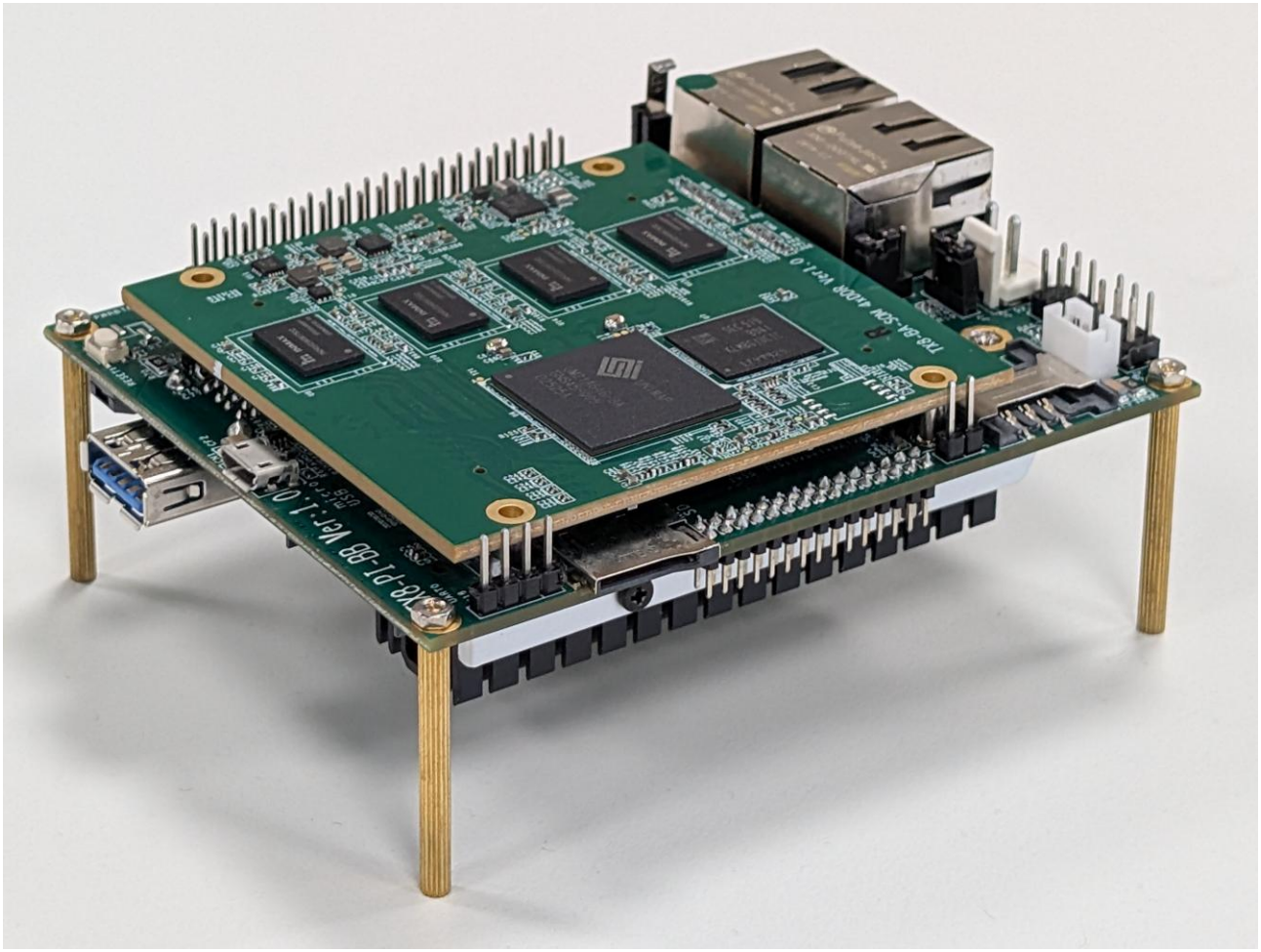
UM-PI AI-Integrated Platform for Drone Applications

Technical product introduction for visual inference and live mission data delivery



UM-PI platform with TX8 host board and M.2 AI acceleration module

MADE IN TAIWAN

English edition | September 2026

PRODUCT POSITION

Executive overview

UM-PI adds customer-defined visual inference to a drone. The ground control machine handles display, tracking, telemetry fusion and operator workflows.

The platform combines a Linux host, high-speed I/O and a dedicated MemryX AI accelerator in a compact board assembly. It is designed for drone OEMs and operators that want to evaluate onboard object detection without moving the entire ground-station software stack onto the aircraft.

Customer need	UM-PI contribution	System boundary
Recognize mission-specific objects	Onboard model preprocessing and inference	Customer defines and validates the mission model
Transmit live imagery and detections	Video forwarding plus class, confidence and box metadata	The data link and ground-station application belong to the complete drone system
Keep operator workflows on the ground	Decoupled inference and display feeds	Rendering, tracking, GPS fusion, recording and alerts run on the ground control machine

Customer outcome

A joint evaluation establishes whether the selected camera, model, data link and ground-control workflow meet the customer's mission requirements. Reference performance values describe the supplied demo configuration and are not a universal guarantee.

DUAL USE APPLICATIONS

Mission scope

The platform supports non-weaponized military and civil monitoring applications in which operators need live imagery and machine-generated detections.

Application family	Representative tasks	Typical operator output
Defense and security	Perimeter monitoring, reconnaissance imagery, vehicle or person classification	Live video, detection overlays and operator decision support
Infrastructure	Power line, bridge, facility and pipeline inspection	Detected objects or anomalies linked to the inspection record
Public safety	Search support, traffic observation and scene monitoring	Prioritized detections for operator review
Commercial aerial analysis	Site survey, asset inventory and recurring visual checks	Structured metadata alongside imagery

The end customer decides the target classes, training data, operating rules and mission decisions. UM-PI provides an AI-integrated platform rather than a formal identification-friend-or-foe system or autonomous engagement function.

RESPONSIBILITIES

Platform scope and system boundaries



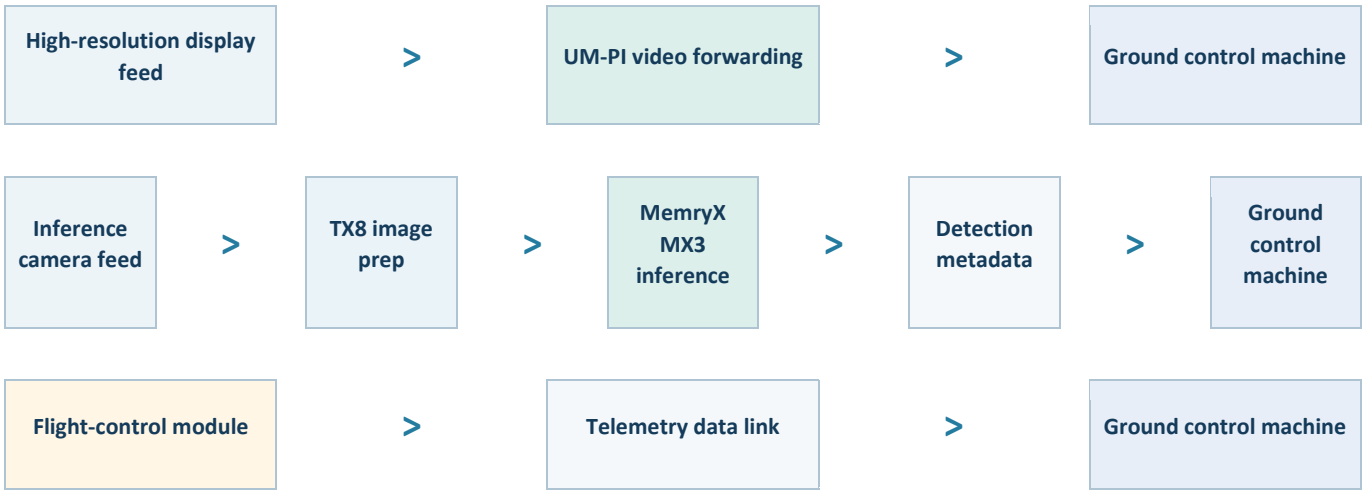
Primary mission-data path

UM-PI performs	Ground control machine performs
Camera acquisition for inference	Display image decoding and rendering
AI input preprocessing	Bounding-box overlay and user interface
MemryX model inference	Multi-frame tracking and tracking IDs
Detection metadata generation	GPS and flight-telemetry fusion
Video and metadata transport handoff	Recording, alert rules and operator workflow

A separate flight-control module supplies navigation, aircraft state and timing information. The ground control machine combines that telemetry with the UM-PI video and detection stream.

BLOCK DIAGRAM

System architecture



Independent video, detection and flight-telemetry paths converge at the ground control machine

The reference design keeps the display stream independent from the inference rate. This lets the ground station receive smooth video while the AI pipeline runs at the rate appropriate for the selected model and resource budget.

RESOURCE ALLOCATION

Processing partition

UM-PI dedicates onboard resources to AI inference and transport preparation. The ground system performs image presentation and mission-level processing.

Stage	Onboard	Ground
Capture	Receive camera feeds	Receive network stream
Preprocessing	Decode the inference feed in software when required, resize and format model input	Decode the display feed
Inference	Run the compiled model on the MemryX accelerator	Optional downstream analytics
Visualization	No mission UI rendering	Draw overlays and present operator controls
Fusion	Forward detections	Combine tracking, GPS, attitude, time and alerts

The current reference code uses CPU-based preparation for the inference stream and MJPEG pass-through for the high-resolution display stream when the camera output is suitable. The design does not depend on onboard hardware video decoding for the operator display path.

PLATFORM COMPOSITION

UM-PI hardware architecture



Layer	Confirmed platform elements
Host compute	UM21A646 with four ARM Cortex-A53 cores at up to 1.2 GHz
Memory and storage	2 GB DRAM and 8 GB eMMC on the SOM
AI expansion	M.2 M-key slot using PCIe Gen3 x4 for the accelerator module
Communications	Two Gigabit Ethernet ports, USB 3.1, USB 2.0, UART and CAN FD
Expansion	40-pin I/O with GPIO, SPI, I2C, UART, ADC and timer functions

The carrier board measures 80 x 100 mm. The SOM measures 68 x 58 mm. System weight, flight power budget, enclosure, cooling and environmental qualification require evaluation in the target airframe.

HOST RESPONSIBILITIES

TX8 host platform

The TX8 host coordinates cameras, model input preparation, accelerator I/O and network transport.

Category	Source-confirmed capability	Role in the reference application
CPU	4 x ARM Cortex-A53, 1.2 GHz	Capture, software preprocessing, metadata handling and orchestration
Memory	2 GB DRAM	Application buffers and runtime data
Storage	8 GB eMMC plus microSD support	Linux image, application and compiled model files
High-speed I/O	PCIe Gen3 x4, USB 3.1 and dual GbE	Accelerator, cameras and network integration
Operating environment	Ubuntu 22.04 and Linux kernel 5.15 in the supplied manual	Native C++ application and device drivers

The source datasheet lists a 3D graphics engine, but this product architecture does not assign display decoding or ground-station rendering to TX8. Those tasks remain on the ground control machine.

CASCADE 100M MODULE

MemryX AI acceleration

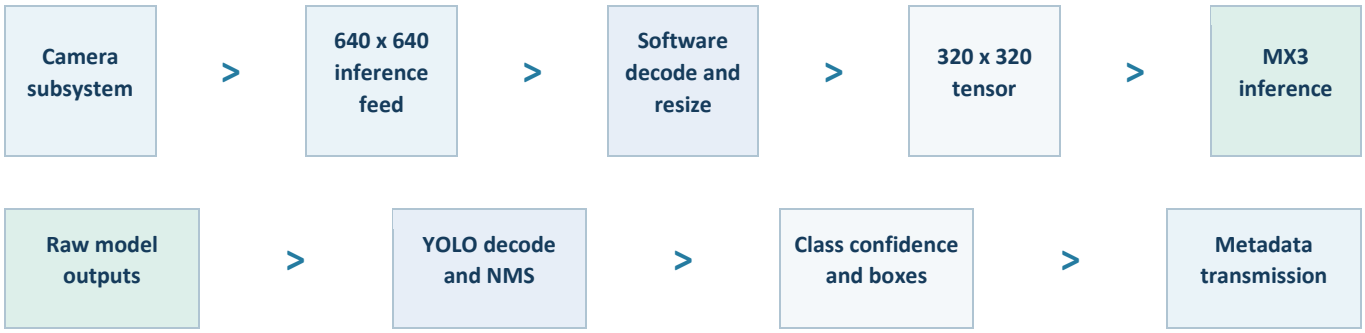
The M.2 accelerator offloads neural-network inference from the TX8 CPU so the host can concentrate on data movement and application control.

Official module specification	Cascade 100M
Accelerators	4 x MemryX MX3
AI performance	25 TFLOPS
On-chip weight storage	42 MB
Module power	1 W to 12 W
Form factor	M.2 2280 M-key, 22 x 80 mm
Host interface	PCI Express Gen 3 and above
Frameworks listed by MemryX	ONNX, TensorFlow, Keras and TFLite

These values are MemryX module specifications, not measured full-system values for a completed drone. The final model, input shape, compiler configuration and thermal design determine application performance.

REFERENCE PIPELINE

Camera and inference data flow



Reference YOLOv10n processing path

The supplied demo uses a precompiled YOLOv10n DFP model with 320 x 320 input and COCO labels. The example demonstrates the software path only. Customers develop and validate models for their own targets, using the MemryX optimized model library as a starting point where applicable.

MEASURED CONFIGURATION

Reference demo configuration

Element	Reference configuration	Interpretation
Display feed	1920 x 1080 MJPEG, requested at up to 60 fps	Passed through without re-encoding when the camera stream is compatible
Inference feed	640 x 640 camera stream	Prepared as 320 x 320 model input
Model	YOLOv10n, 80 COCO classes	Example only; customer mission model is separate
Inference modes	Every frame or software-throttled to approximately 30 fps	Balances inference load with other system work
Host allocation	Four cores or a two-core affinity option	Demonstrates resource allocation choices
Viewer	Browser over HTTP and WebSocket on port 8081	Controlled-LAN demonstration, not a production security design

The project README describes approximately 30 to 60 fps inference modes. Actual throughput and end-to-end latency depend on the model, camera, input format, data link and integration choices.

REFERENCE IMPLEMENTATION

Software stack



Component	Function
MemryX Runtime	Loads the compiled DFP and executes accelerator input and output callbacks
OpenCV and TurboJPEG	Camera acquisition, software image preparation and fallback image transcoding
Reference C++ application	Coordinates capture, inference, YOLO output decoding and metadata publishing
Embedded HTTP and WebSocket server	Forwards the display image and detection JSON to a browser
Browser viewer	Draws boxes, smooths short-term detections and shows display and inference rates

The source package is reference code for evaluation. Customers remain responsible for production software architecture, mission behavior, fault handling and qualification.

DEVELOPMENT FLOW

Customer model workflow



Party	Primary contribution
Single solution provider	One customer-facing interface for platform evaluation, accelerator integration and technical support
Unimap technology	UM-PI hardware platform, board software and platform engineering inputs
MemryX technology	Accelerator runtime, compiler tools, documentation and optimized model resources
Customer or system integrator	Mission model, datasets, camera selection, flight-control integration, ground software and qualification

DATA CONTRACT

Detection data and ground integration

Data item	Current demo	Evaluation or production extension
Video frame	JPEG payload	Selected transport, compression and reliability policy
Frame identity	Sequence number, width and height	Timestamp synchronization across aircraft systems
Detection	x, y, width, height, class and confidence	Mission schema, filtering and audit fields
Tracking	Short-term browser smoothing without persistent IDs	Ground-side tracker and persistent tracking IDs
Position	Not included	GPS, attitude and altitude from the flight-control module
Alerting	Not included	Ground-side rules, severity and operator acknowledgement

The evaluation should define synchronization, coordinate conventions, loss behavior and metadata versioning before the ground-control application consumes the stream.

REFERENCE OBSERVATIONS

Performance and resource behavior

The demo separates display delivery from inference so each path can run at an appropriate rate.

Design choice	Effect
Display and inference feeds are separate	Display frame rate does not have to equal inference frame rate
Latest-wins frame handling	Avoids growing queues when a consumer falls behind
MJPEG pass-through	Avoids display-stream decode and re-encode on TX8 when the camera stream is compatible
Inference throttling	Reduces CPU, accelerator and memory traffic when full-rate inference is unnecessary
CPU affinity options	Reserves host cores for other application work during evaluation

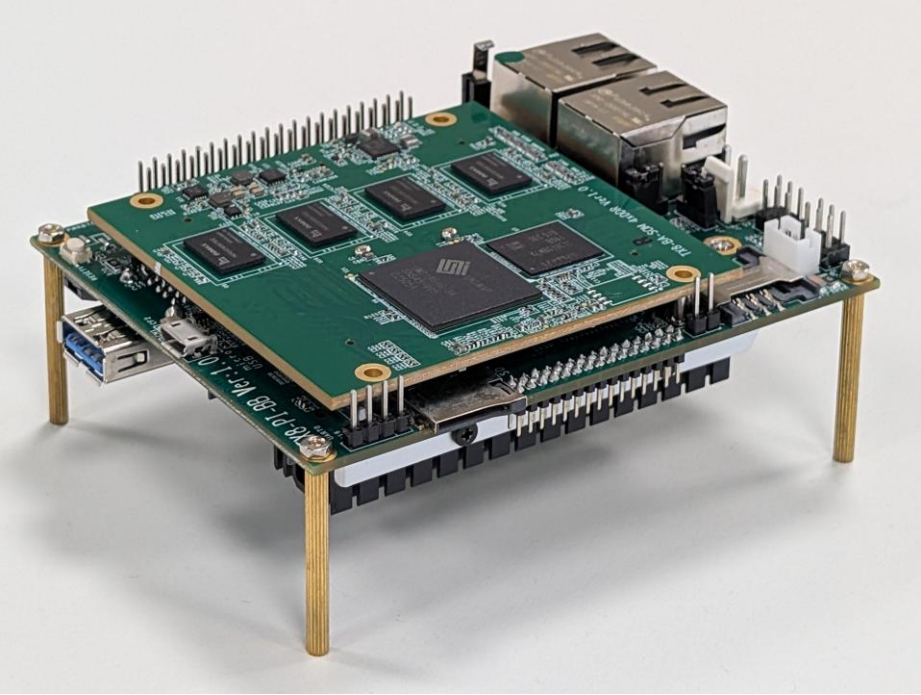
The reference values are configuration-specific. Customer evaluation should measure the selected model and camera in the target airframe and data-link environment.

INTEGRATION DATA

Interfaces and mechanical baseline

Item	Confirmed value
Carrier board	80 x 100 mm
SOM	68 x 58 mm
Development power input	5 V / 6 A supply specified for development setup
Camera interface	USB UVC in the reference system
Network	Two Gigabit Ethernet ports on the UM-PI carrier
Expansion	M.2 PCIe Gen3 x4, mPCIe, USB, CAN FD, UART, SPI, I2C and GPIO

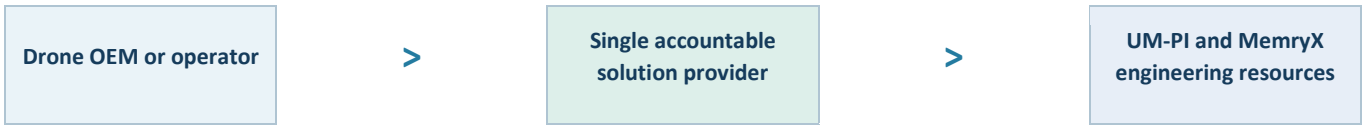
The development supply rating does not establish flight power consumption. The target design must confirm complete-system power, mass, cooling, vibration, EMI/EMC, enclosure and environmental limits.



UM-PI reference hardware assembly

SINGLE INTERFACE

Customer support model



The customer works through one solution provider for evaluation and integration support.

- Product fit review covering the camera, model, data link and ground-control architecture
- Coordination of UM-PI platform questions and MemryX accelerator topics
- Reference code and model-workflow guidance for customer development
- Joint technical evaluation before production commitments

The customer retains ownership of mission requirements, datasets, application software, security policy and system qualification.

ENGINEERING BOUNDARY

Reference demo and production deployment

Area	Reference demo	Production deployment
Network	Controlled local network	Customer-approved aircraft data link and network architecture
Transport security	HTTP and WebSocket without authentication	Authenticated and encrypted transport, access control and key management
Resilience	Automatic application restart after a dropped stream	Defined fault detection, recovery, logging and health monitoring
Data fusion	Video and detection metadata	Synchronized telemetry, tracking, geotagging and mission records
Qualification	Engineering demonstration	Airframe, environmental, EMI/EMC, cybersecurity and operational validation

The supplied software demonstrates the integration concept. The solution provider and customer jointly define the evaluation needed for the intended operating environment.

NEXT STEP

Evaluation path



Evaluation input	Purpose
Target classes and representative imagery	Confirm the model and dataset scope
Camera outputs and physical interface	Confirm display and inference feeds
Ground-control software and data link	Define video, metadata and telemetry integration
Airframe constraints	Confirm board placement, power and thermal design
Operational acceptance criteria	Define evidence before any production commitment

Recommended next action

Start a customer-specific technical evaluation with the single solution provider. The evaluation should use the customer's target model, camera, ground-control workflow and aircraft constraints.

Illustration prompt library

The following prompts support later image generation. They should produce neutral, non-weaponized visuals that match the white-background technical design of this document.

Cover scene

Professional product visualization of a compact quadcopter used for aerial monitoring, UM-PI edge AI module visible as an integrated electronics payload, clean white studio background, subtle navy and teal accents, realistic engineering style, no weapons, no logos, ample negative space on the left for a title, 16:9 composition.

Mission operation

High-altitude drone observing a remote facility and surrounding roads, subtle computer-vision bounding boxes around vehicles and people, ground control operator shown in a separate lower-right inset, white and pale gray technical background, restrained navy and teal palette, non-weaponized, realistic, 16:9.

System data flow

Isometric technical illustration showing a drone camera, onboard edge AI computer, secure data link, separate flight controller and ground control workstation, clearly separated video, detection metadata and telemetry paths, white background, minimal blue and teal engineering aesthetic, no decorative text, 16:9.

Evaluation setup

Engineering lab evaluation of a drone edge AI platform connected to a USB camera and a ground control laptop, clean bench, white background, realistic professional photography, blue and teal accents, no people required, no visible brand marks, 16:9.

SOURCE TRACEABILITY

References and contact

Primary sources

- Unimap Technology, UM-PI AI System User Manual, Version 1.0, August 2026.
- Unimap Technology, UM2xA646 Datasheet, document UM2xA646-DS, Revision 3, 13 May 2025.
- UM-PI Real-Time Object Detection Demo README, build and run scripts, C++ source and browser viewer, August 2026.
- UM-PI Real-Time Object Detection Demo Operation Manual, Version 3, August 2026.
- MemryX, Cascade 100M M.2 AI Accelerator Module product page, accessed September 2026, https://memryx.ai/cascade-100m-m-2-ai-accelerator-module/.

Customer contact

Field	Customer-facing information
Solution provider	[Solution Provider]
Contact	[Name and title]
Email	[Email address]
Website	[Website]

UM-PI by Unimap Technology | Made in Taiwan